

CS 6603 – Homework Project: AI/ML

Assignment – Part 1 Report

Ndubusi Anyanwu

Introduction

This report: (1) classifies subgroups into protected classes, (2) computes correlations using compact encodings, (3) analyzes population statistics and samples, and (4–6) studies a chosen protected class (sexual_orientation) and its subgroups.

Step 0

I removed two misaligned rows (including the blank 52671 entry) that contained incorrect or missing Wiki_ID and toxicity values to restore proper dataset alignment.

Step 1: Protected classes & subgroups

Step 1- Protected Classes & Subgroups	
Protected Class	Subgroups
religion	buddhist, christian, jewish, muslim
gender	female, male, nonbinary, transgender
sexual_orientation	bisexual, gay, heterosexual, lesbian, lgbtq, queer, straight
race_ethnicity	african, asian, black, hispanic, indian, latino, white
nationality	american, canadian, chinese, japanese, mexican
disability	blind, deaf
age	elderly, old, older, teenage, young, younger
misc	african american, catholic, european, homosexual, latina, latinx, lgbt, middle aged, middle eastern, millennial, paralyzed, protestant, sikh, taoist, trans

It is to be noted that some subgroups can belong to more than one protected class and would be dual-class. In this report I keep it under nationality only so that I can avoid double-counting, but either choice is acceptable if applied consistently. For instance I place chinese under nationality and not race or ethnicity to avoid double-counting.

Step 2: Calculating toxicity correlations

The table compacts each protected class into one ordinal coded column and reports the Pearson correlation with toxicity

2.1

I removed any row where all subgroup flags were FALSE, keeping only comments that mention at least one identity attribute.

Reduction summary

Rule	Result
Keep a row if any subgroup column is TRUE	Retained
Drop a row if all subgroup columns are FALSE	Removed
Size of reduced dataset (n)	75,699

2.2 Objective numerical scheme

For each protected class, I created a single ordinal code per row:

FALSE → 0

If one or more subgroups are TRUE, I assigned the max code among those subgroups.

Subgroup codes were set by sorting subgroups by their population mean toxicity lowest mean = 1, next = 2, and so on.

This was to yield a reproducible, objective ordering that reflects observed toxicity levels rather than subjective choices.

Table 2-B. Example code map (sexual_orientation)

Subgroup	Code
heterosexual	1
lgbtq	2
straight	3
gay	4
lesbian	5
bisexual	6
queer	7

2.3 Compacted dataset

I collapsed all subgroup columns for each protected class into one ordinal column and those compact columns were then correlated with TOXICITY.

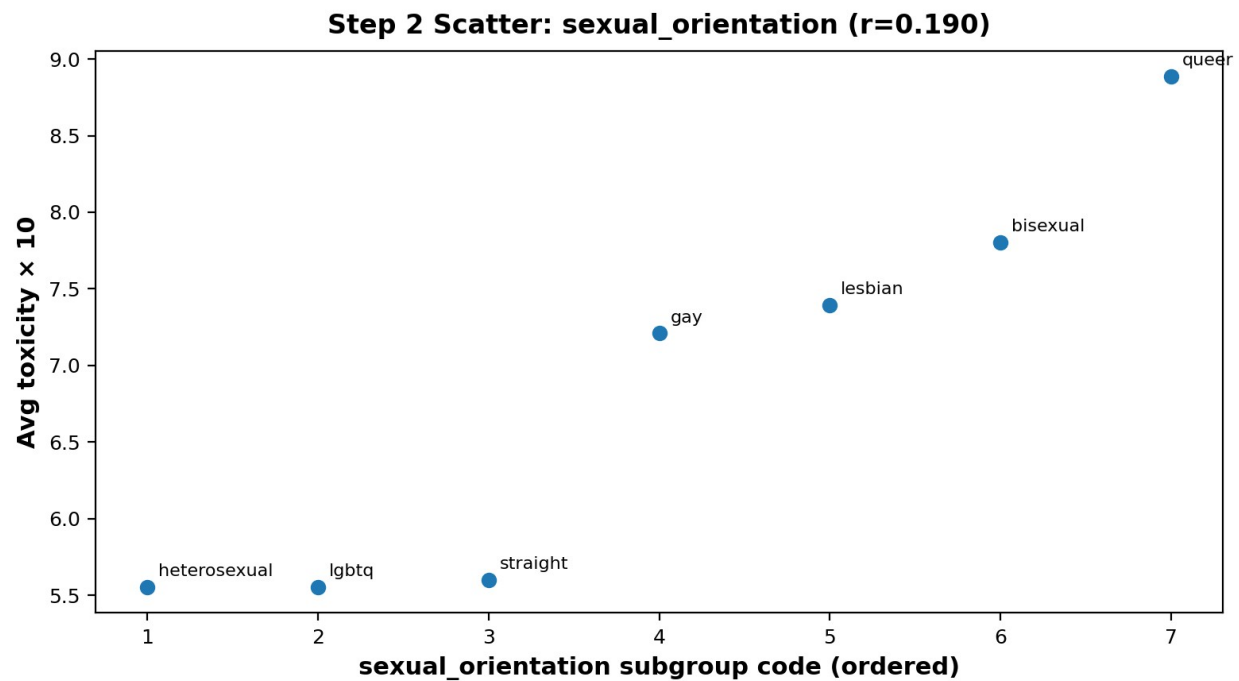
Table 2-C. Compacted columns

Protected class	New compact column
religion	religion_compact
gender	gender_compact
sexual_orientation	sexual_orientation_compact
race_ethnicity	race_ethnicity_compact

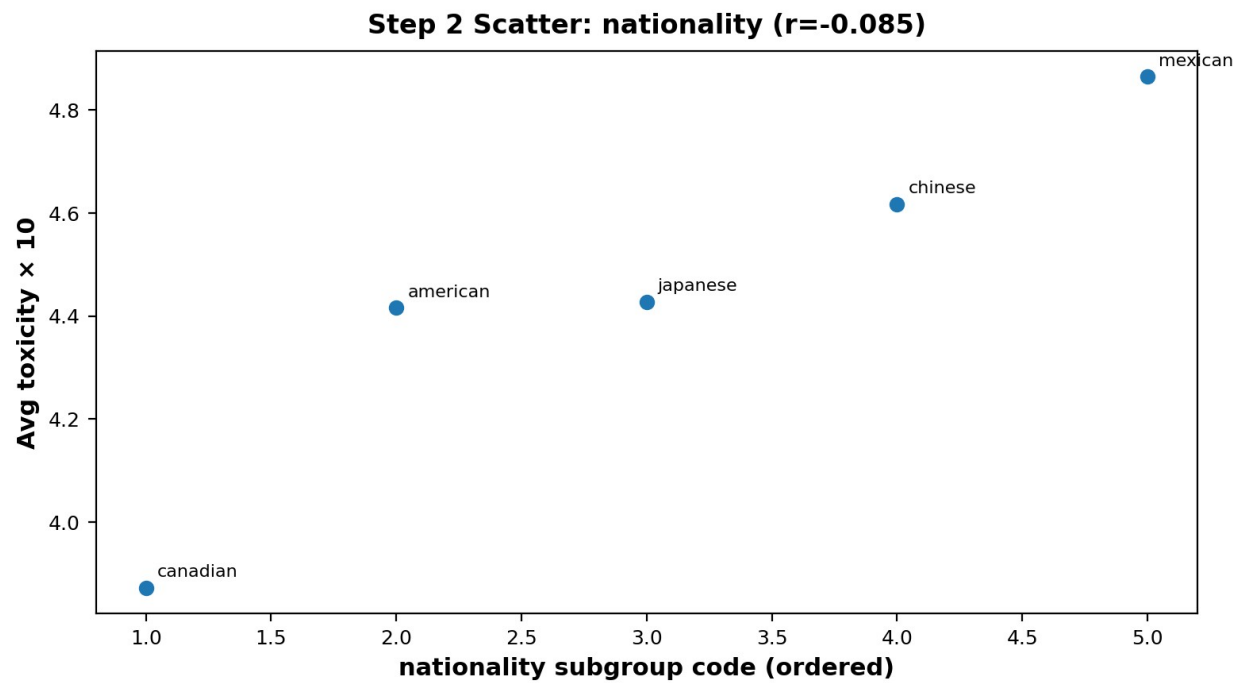
nationality	nationality_compact
disability	disability_compact
age	age_compact

Correlations between compacted protected-class columns and TOXICITY			
Protected Class	Correlation r	r	Strength
sexual_orientation	0.19021771695127135	0.19021771695127135	very weak
nationality	-0.08519784347571134	0.08519784347571134	very weak
gender	0.0737912964470454	0.0737912964470454	very weak
misc	0.04822056837059826	0.04822056837059826	very weak
disability	0.031815138323115612	0.031815138323115612	very weak
religion	-0.01615956001696324	0.016159560016963243	very weak
age	0.00447954125661837	0.00447954125661837	very weak
race_ethnicity	0.00387379808078663	0.00387379808078663	very weak

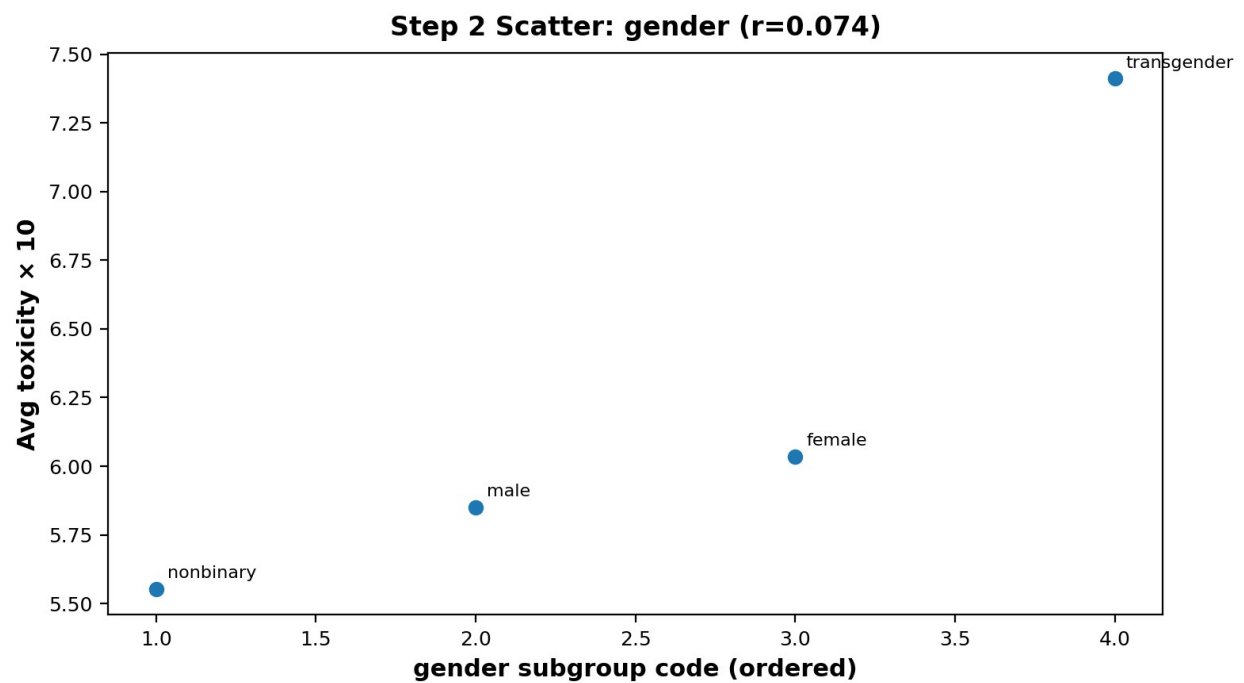
Scatter plots for the three largest |r| values:



. Step 2 scatter — sexual_orientation



Step 2 scatter — nationality



. Step 2 scatter — gender

Step 2 — Direct Q&A

Q: Do you agree with the correlation values? Why or why not?

A: Yes I agree with the correlation values; the very weak correlations in this case for example, sexual_orientation $r \approx 0.19$, match the scatter plots, which show only slight monotonic trends. The dataset is very noisy and also multi factorial, so I expect small r-values even when subgroup means may differ.

Step 3: Analyzing toxicity on the reduced dataset

Step 3 - Overall stats & samples

Dataset	n	Mean	Std Dev	95% Range	MoE (95%)
Population (Reduced)	75699	0.5500964421351405	0.36184945947695357	[-0.174, 1.274]	nan
10% Sample	7570	0.5501062127969616	0.36227556906964437	nan	0.008161071427772606
60% Sample	45419	0.5495415447922014	0.36212242369969333	nan	0.0033303716872933346

Step 4: Chosen protected class — sexual_orientation

Below I compute the mean and standard deviation for the protected class in population and then the samples. Then I check whether sample means fall within the population MoE.

Step 4 - Stats (sexual_orientation)

Chosen Class	Dataset	n	Mean	Std Dev	MoE (95%)
sexual_orientation	Population (class present)	10598.000	0.686	0.311	
sexual_orientation	10% Sample (class present)	1073.000	0.693	0.310	0.019
sexual_orientation	60% Sample (class present)	6350.000	0.689	0.311	0.008

Step 4 — Stats for chosen class

Step 4 - MoE checks (sexual_orientation)

Dataset	Within population $\pm$ MoE?	Details
10% Sample	No	0.550 ± 0.008 vs 0.693
60% Sample	No	0.550 ± 0.003 vs 0.689

Step 4 — MoE checks

Step 4 — Direct Q&A

Q: 10% dataset — Does the protected-class sample mean lie within the population MoE? Explain.

A: No, instead the 10% sample mean for sexual_orientation falls outside 0.55 ± 0.008 , which is plausible due to the given higher variance and limited representation of several subgroups within this small sample.

Q: 60% dataset — Does the protected-class sample mean lie within the population MoE? Explain.

A: No and despite being larger, the 60% sample still deviates from 0.55 ± 0.003 , which would indicate the persistent subgroup imbalance rather than pure sampling noise.

Step 5: Subgroups of the chosen protected class

The following tables report subgroup level mean, standard deviation, and MoE for population and samples, with checks for MoE inclusion.

Step 5 — Population

Step 5 - Subgroup stats (population, sexual_orientation)

Subgroup	n	Mean	Std Dev	MoE (95%)
queer	1514	0.8884331742701453	0.12306254116199027	0.006198962548257067
bisexual	1514	0.7804491662965654	0.23400001468853343	0.01178715564987739
lesbian	1514	0.7395405172780714	0.2560950762238464	0.012900138184331542
gay	1514	0.7209887482813739	0.274391053190778	0.01382175149518046
straight	1514	0.5601547028124174	0.35384051196236	0.01782381593131164
heterosexual	1514	0.5551644657826948	0.3444332883044277	0.017349950963239522
lgbtq	1514	0.5551644657826948	0.3444332883044277	0.017349950963239522

Step 5 — 10% sample

Step 5 - Subgroup stats (10%, sexual_orientation)

Subgroup	n	Mean	Std Dev	MoE (95%)
queer	158	0.8857454927784809	0.13933925400929956	0.02172706226574727
bisexual	160	0.7911333760250001	0.22794021830601202	0.03531970274994726
gay	152	0.7377492332960527	0.27041422028319406	0.04298963952614068
lesbian	149	0.7375516543758389	0.24613571872408338	0.03952187856149838
lgbtq	167	0.5844775681736527	0.3508444498423938	0.0532123509135656
straight	151	0.5802215526953641	0.35331135252650936	0.056354057403624765
heterosexual	136	0.5155814572720587	0.3461176584406754	0.05817151494110568

Step 5 — 60% sample

Step 5 - Subgroup stats (60%, sexual_orientation)

Subgroup	n	Mean	Std Dev	MoE (95%)
queer	948	0.8892543460052742	0.12213534464492874	0.007774872306069778
bisexual	906	0.7862132973532009	0.232011636542598	0.01510781784832926
lesbian	907	0.7393479201356119	0.25745155128828706	0.016755135365199292
gay	910	0.7335422743989012	0.2677507381866991	0.01739666706372181
straight	893	0.5564499229540874	0.3573328354345049	0.023437067316375845
lgbtq	875	0.552386192776	0.3454611710217626	0.022890289516951007
heterosexual	911	0.5490207692195389	0.3441284575080923	0.022346909526280396

Step 5 — 10% sample — MoE checks

Step 5 - MoE checks (10%, sexual_orientation)

Dataset	Subgroup	Within population $\pm$ MoE?	Sample MoE
10% Sample	queer	No	0.008161071427772606
10% Sample	bisexual	No	0.008161071427772606
10% Sample	gay	No	0.008161071427772606
10% Sample	lesbian	No	0.008161071427772606
10% Sample	lgbtq	No	0.008161071427772606
10% Sample	straight	No	0.008161071427772606
10% Sample	heterosexual	No	0.008161071427772606

Step 5 — 60% sample — MoE checks

Step 5 - MoE checks (60%, sexual_orientation)

Dataset	Subgroup	Within population $\pm$ MoE?	Sample MoE
60% Sample	queer	No	0.0033303716872933346
60% Sample	bisexual	No	0.0033303716872933346
60% Sample	lesbian	No	0.0033303716872933346
60% Sample	gay	No	0.0033303716872933346
60% Sample	straight	No	0.0033303716872933346
60% Sample	lgbtq	Yes	0.0033303716872933346
60% Sample	heterosexual	Yes	0.0033303716872933346

Step 5 — Direct Q&A

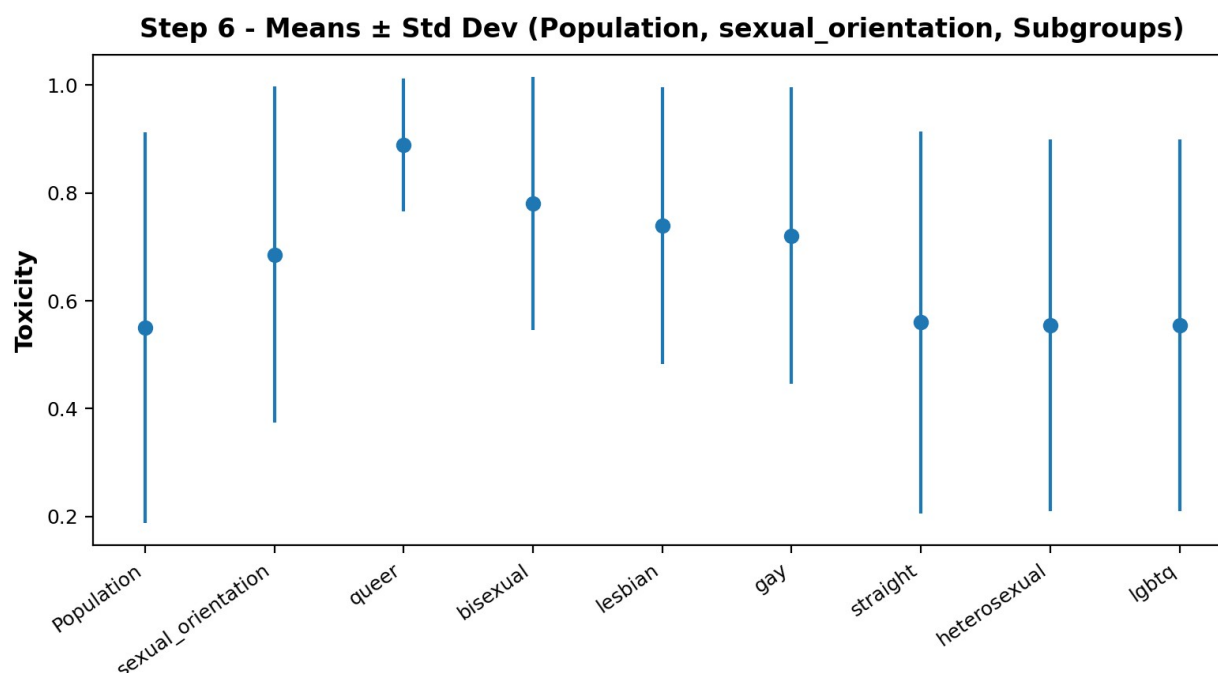
Q: 10% dataset — Do subgroup means lie within the population MoE? Explain.

A: No for all subgroups and due to the small n per subgroup and relatively large standard deviations, their means naturally fell outside of the tight population interval.

Q: 60% dataset — Do subgroup means lie within the population MoE? Explain.

A: My answer is mixed; for the heterosexual and lgbtq the answer would be Yes and for others = No. Larger n stabilizes estimates for more common labels, but high mean groups for example queer and bisexual remain outside due to systematic shifts and not sampling error.

Step 6: Plots and toxicity analysis



Step 6 — Direct Q&A

Q: Which subgroup has the highest TOXICITY value? Justify.

A: Queer has the highest toxicity with a mean of ~0.89, it is clearly above both the protected-class and population means on the combined error bar chart.

Q: Which subgroup has the lowest TOXICITY value? Justify.

A: Heterosexual as well as the broader lgbtq tag at ~0.55 have the lowest toxicity value. They have a means that align with or slightly below the population mean and also have overlapping error bars.

Q: Which subgroup differs most from the population mean? Justify.

A: The subgroup that differs the most would be Queer. This gap between its mean of ~0.89 and the population mean which is ~0.55 is largest and exceed more than one population standard deviation.

Q: What type of human bias is in the data; how to quantify; how to minimize?

A: The results point mainly to representational or labeling bias. I quantify it with gaps between subgroup means, small-but-directional class-toxicity correlations, and fairness metrics; to reduce it. What would also help would be to add counterfactual balanced examples, reweight, or calibrate thresholds by subgroup, additionally when training a model use adversarial debiasing.

Reflection

Toxicity varies a lot across identity subgroups even when the overall correlations look weak, which means some real effects are getting averaged away. In our data, groups like queer and bisexual sit consistently above the population mean, while heterosexual and the broader LGBTQ tag cluster near it but with large variance. That pattern points to representational/labeling bias and long-tail sampling issues—something we can see more clearly by comparing subgroup mean gaps and using MoE checks to separate noise from systematic shifts. Going forward, I'd rebalance with counterfactual augmentation and apply subgroup-aware calibration/thresholding to reduce downstream harms.